

ASHUTOSH ROY

Software Engineer

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SUMMARY

Software Engineer with **1+ year of hands-on experience** building scalable backend applications, REST APIs, and AI-powered solutions. Experienced through internships at **IIT Delhi** and software startups. Skilled in **Python, FastAPI, SQL, Docker, Git, and Linux**, with experience improving API performance, building production-ready services, and developing reliable software for real-world applications.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built and deployed **FastAPI** REST APIs for search and document retrieval, supporting **1,000+** daily requests with low response time.
- Developed a Python OCR pipeline using Tesseract, Marker, and Gemini OCR to process **10,000+** documents, reducing manual processing and improving data quality.
- Built a hybrid search system using **FAISS, Pinecone, and Apache Solr**, improving search accuracy by **35%**.
github.com/ashutoshroy02/TIE

Software Engineer Intern

Nov 2024 – Dec 2024

LeanTactics Solutions Pvt. Ltd.

Remote

- Built backend services using **Python** and **FastAPI** for scalable AI applications with reliable performance.
- Integrated ML inference APIs into production systems, enabling fast and seamless real-time predictions.
- Optimized API performance, reducing response latency by **30%** and improving overall system efficiency.

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned Mistral and LLaMA models for healthcare applications, improving medical summarization quality and chatbot performance.
- Built an AI chatbot that reduced patient query resolution time by **40%**, improved customer experience, and lowered support costs.
- Built end-to-end ML pipelines from data preprocessing to API deployment, reducing deployment time and speeding up AI solution delivery.

PROJECTS

URL Shortener | Python - FastAPI - PostgreSQL - Redis - Docker

2026

- Built a scalable URL shortener with REST APIs, custom short links, and analytics using FastAPI and PostgreSQL.
- Integrated Redis caching to reduce response time by **40%** and improve system performance.

InterviewPrep Platform | Python - FastAPI - Astro Engine - SQLite - Git

2025

- Built a full-stack interview preparation platform with coding questions, quizzes, mock interviews, and progress tracking.
- Designed backend APIs to improve performance and deliver a smooth user experience.

Exam-Helper RAG System | Python - LLaMA 3 - FastAPI - LangChain - FAISS - Pinecone

Dec 2024 – Apr 2025

- Built a RAG-powered question answering system with support for PDFs, images, and YouTube transcripts.
- Implemented FAISS and Pinecone retrieval to improve answer accuracy and enable fast semantic search.

TECHNICAL SKILLS

Languages: Python, C++, SQL, DSA

Backend Development: FastAPI, REST APIs, Authentication, API Integration, Microservices, Multithreading, Linux

Databases: MySQL, PostgreSQL, Redis, FAISS, Pinecone

System Design: Object-Oriented Programming (OOP), Low-Level Design (LLD), Caching, Load Balancing, Scalability, Database Design

Cloud & DevOps: Docker, GitHub Actions (CI/CD), AWS (EC2, S3)

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, Streamlit

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

ACHIEVEMENTS

- Published **2 research papers**, including one accepted at **RECCAP 2026 (IEEE)** and another co-authored with **IIT Delhi**.
- Solved **150+** DSA problems on LeetCode, strengthening data structures and algorithms.
- Won **2 AI/ML Hackathons**, building innovative AI solutions under time constraints.
- Made contributions to **open-source AI/ML projects** on GitHub and Hugging Face.
- Earned certifications in **Machine Learning, Deep Learning, Generative AI, NLP, and Python**.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana